

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Col - A12

Network Design for a 20-person startup - star topology.

A star topology is ideal for a 20-person startup because all devices are connected to a central switch. This makes the network easy to manage, monitor, and trouble shoot. A managed layer 2/layer 3 switch with pro. at least 24 ports can be used to connect workstations, printers, and VoIP phones while providing room for future expansion.

The major advantage of this design is fault isolation.

If the cable or network connection of one workstation fails, only that workstation is affected. The other 19 employees can continue using the network. Centralized cable management also makes it easier to add or relocate devices.

- central switch provides simple management and troubleshooting.

- Cat6 UTP supports high-speed ethernet within the office.

Network Design for a Bank - Mesh Topology.

A bank requires very high availability and minimum downtime. Therefore, a full or partial mesh topology is suitable in a full mesh providing alternative communication paths. A partial mesh can be used at the core to reduce cost while still providing redundancy.

Network Design for a 20-person startup - star topology.

A star topology is ideal for a 20-person startup because all devices are connected to a central switch. This makes the network easy to manage, monitor, and trouble shoot. A managed layer 2/layer 3 switch with pro. at least 24 ports can be used to connect workstations, printers, and VoIP phones while providing room for future expansion.

The major advantage of this design is fault isolation.

If the cable or network connection of one workstation fails, only that workstation is affected. The other 19 employees can continue using the network. Centralized cable management also makes it easier to add or relocate devices.

- central switch provides simple management and troubleshooting.

- Cat6 UTP supports high-speed ethernet within the office.

Network Design for a Bank - Mesh topology.

A bank requires very high availability and minimum downtime. Therefore, a full or partial mesh topology is suitable in a full mesh, providing alternative communication paths. A partial mesh can be used at the core to reduce cost while still providing redundancy.

The main advantage of a mesh design is fault tolerance. If one link fails, traffic can use another available path. Routing protocols such as OSPF or BGP can help reroute traffic automatically. Redundant power supplies, UPS system, and RAID storage can further improve reliability.

- Mesh provides multiple paths and high network availability.
- Fiber supports high speed and reliable communication.

3. Network Design for a small classlab - Bus Topology.

For a small classroom laboratory with a limited budget, a bus topology can provide a low-cost networking solution. In a traditional bus topology, all computers share a common communication line, reducing the requirement for a central networking device.

The major limitations of a bus topology is that all computers share the same communication medium. This can lead to collisions and reduced performance. Also, a ~~break~~ in the main cable can affect all connected computers.

- Bus topology requires less hardware and can reduce cost.
- Computers share a common communication medium.
- A backbone failure can affect the entire network.

Network design for a university campus - Hybrid Topology.

A large university campus contains thousands of users distributed across lecture halls, laboratories, libraries, hostels, administrative building. A single topology is not suitable network design is appropriate.

The design follows a hierarchical structure consisting of core, distribution, and access layers. It makes the network easier to manage and allows new buildings or departments to be added without major change to the existing infrastructure.

- combined ring/mesh and star structure according to requirements.
- Fibre backbone provides high speed and redundancy.
- Hierarchical design makes expansion easier.

5. network design for a company with multiple departments.

A company with departments such as HR, Finance, Sales, and IT requires a structured and secure network. Each department can use a dedicated layer 2 access switch to connect computers, printers and other devices.

This hierarchical design provides good performances, security, reliability, and scalability. New users, departments or devices can be added without redesigning the entire network.

- VLAN's separate departments and improve network security.
- Layer 3 routing enables controlled communication b/w departments.
- Hierarchical design supports reliability and future expansion.